

5 (a) (i) Waves from loudspeaker travel down tube and are reflected at closed end

B1

Two waves travelling in opposite directions with same frequency and wavelength overlap

B1

(ii) 0.51 m or 0.85 m

A1

(b) (i) number = 5

A1

(ii) phase difference = 180° or π rad (must have unit)

A1

(iii) $T = \frac{1}{50} = 20$ ms

C1

1. reflection of curve about the line XY and labelled M

A1

2. horizontal line between X and Y and labelled N

A1

Comments: Many students are unable to obtain the correct phase difference or draw N correctly.

6 (a) p.d. per unit length = $V_{\text{cell}} / 100 = 2.0 / 100 = 0.020 \text{ V cm}^{-1}$

C1